

Certification Program in Applied Data Science and Machine Learning

Day	Main Theme	Detailed Topics
Day 1	Exploratory Data Analysis	• Introduction to Data Science and AI/ML ecosystem • Python basic library for data science • Introduction to NumPy and Pandas • Data cleaning and preprocessing • Hands-on: Data cleaning using Pandas • Importance of statistics in machine learning • Mean, median, mode, variance, standard deviation • Correlation and covariance • Data distribution and skewness • Data visualization using Matplotlib and Seaborn: Bar chart, line chart, histogram, box plot, scatter plot • Feature scaling and normalization • Encoding categorical variables • Hands-on: Exploratory Data Analysis on a real dataset
Day 2	Supervised Learning Introduction, Regression	• Introduction to machine learning • Types of machine learning: supervised, unsupervised, semi-supervised • Understanding input features and target variable • Train-test split concept • Regression problem overview • Simple Linear Regression • Multiple Linear Regression • Decision Tree Regression & Random Forest Regression overview • Regression evaluation metrics: MAE, MSE, RMSE, R² score • Hands-on: Regression model building using Scikit-learn • Introduction to PyCaret • PyCaret installation and environment setup • PyCaret regression workflow • Model comparison using PyCaret • Hands-on: Regression model building using PyCaret
Day 3	Supervised Learning: Classification	• Introduction to classification problems • Binary classification vs multi-class classification • Naive Bayes • Decision Tree Classifier • Random Forest Classifier • Data preprocessing for classification • Class imbalance introduction • Classification evaluation: Accuracy, precision, recall, and F1-score • Hands-on: Classification model building using Scikit-learn • PyCaret classification workflow • Model comparison using PyCaret classification module

		• Hands-on: Classification using PyCaret
Day 4	Hyperparameter Tuning, Ensemble Methods & Clustering	• Overfitting and underfitting • Bias-variance trade-off • Cross-validation • Hyperparameter tuning: Grid Search, Random Search • Hands-on: Model Tuning • Introduction to ensemble learning: Bagging, Boosting • Clustering Introduction • K-Means clustering • Elbow method • Silhouette score • Hands-on: K-Means clustering
Day 5	Introduction to Neural Network & Generative AI	• Introduction to neural networks • Biological neuron vs artificial neuron • Perceptron concept • Artificial Neural Network architecture: Input layer, hidden layer, and output layer, Weights, bias, and activation functions • Forward propagation intuition • Loss function and optimization • Backpropagation intuition • Hands-on: Model building with Neural Network • Introduction to Generative AI • Traditional AI vs Generative AI • Large Language Models overview • Prompt engineering basics: Zero-shot and few-shot prompting • Hands On: Using pre-trained models: Text generation, text classification, and summarization